

TERRY DENG

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Education

University of Waterloo

Expected Graduation: 2031

Bachelor of Mathematics (Co-op)

Waterloo, ON

- Courses: MATH 147 Calculus I (Advanced), MATH 145 Algebra (Advanced), CS 135 Designing Functional Programs, DATASC 101 Intro to Data Science, ENGL 119 Communications in Math and CS

Bergen County Academies

Class of 2026

Academy for the Advancement of Science and Technology

Hackensack, NJ

- **GPA: 3.95/4.0** | **SAT: 1580** | Scores of **5** on all AP exams
- Courses: Advanced Math Topics (abstract algebra: groups, rings, fields, intro to Galois theory), Advanced Topics in Physics, Foundations of Computer Science, SQL
- **AP:** Analytical Calculus BC+, Statistics, Microeconomics, Physics C

Technical & Analytical Skills

Programming & Tools: Python, SQL, Java

Analytical: Cost-benefit analysis, causal inference, probability modeling, empirical data analysis

Languages & Interests: Mandarin Chinese (conversational); Piano (RCM Level 10 repertoire, 13+ years); Bass (3 years)

Projects

Genshin Wish Calculator | *Live Site* | *Illustrated Explainer* | *Source Code*

TypeScript | **React**

- Built a browser calculator giving the exact odds of completing a prioritized list of Genshin Impact gacha goals (characters and weapons, with constellation/refinement targets) by every pull count up to a budget
- Modeled banner mechanics (pity, 50/50 guarantees, Capturing Radiance, Epitomized Path) as a **Markov chain** over thousands of states, computing exact probabilities with **transition matrices** rather than Monte Carlo simulation
- Taught myself Markov chains and state-space enumeration through the project; deployed on GitHub Pages, runs entirely in the browser

Work Experience

National Museum of Mathematics (MoMath) | *Integrator (Floor Staff)* | *New York, NY*

Sep 2025 – May 2026

- Rotated between stations every 30 min, answering questions and helping with interactive exhibits; **actively engaged** visitors with prompting questions, communicating math to a general audience
- Attended optional morning trainings on exhibits and broader math topics with **Chaim Goodman-Strauss** and **Art Benjamin**

ACE Education and Consulting Group | *SAT Teaching Assistant, Camp Counselor*

Jun 2024 – Aug 2025

- Assisted SAT Math and Reading/Writing prep (**50%** of students scored **1500+** on the final practice test); created vocabulary quizzes and practice tests; led problem-solving sessions
- Taught math and led science experiments and group activities for **25+ students** in grades 1–6, organized by skill level

Research Experience

PFAS Mitigation Cost-Benefit Analysis for Ridgewood Water

Jun 2024 – Sep 2025

Research, with expert advisor

- Conducted a **cost-benefit analysis** of Ridgewood Water's ongoing PFAS Treatment Plan with an expert advisor
- Applied **PFAS health burden methodologies** to local contamination data, using Ridgewood Water's **financial data**

Economic Impacts of One-Child Policy Enforcement Disparities in China

Sept 2024 – Jun 2025

Research

- Analyzed the **causal relationship** between One-Child Policy enforcement disparities and **economic inequality** in China
- Used **asset and income data** from the **China Health and Nutrition Survey** to quantify economic outcomes

Volunteering & Leadership

150+ Community Service Hours

Schoolhouse.com | *Certified Volunteer SAT Prep Tutor*

Feb 2024 – Jun 2025

- Tutored SAT Math; built custom curricula for **26 students** from practice test results, with lectures and problem-solving sessions

Rockland Centennial Leo's Club | *Membership Chairperson*

Sept 2024 – Jun 2026

- Organized concerts and events, managed performer bookings, and curated programs

Moptica Band | *Bassist*

Sept 2023 – Present

- Performed at charity events, nursing homes, and school assemblies, identifying and securing performance opportunities
- Released original music as a band to **14** streaming platforms